

Year 7 · Chapter 4 · Fractions and percentages: answers (teacher copy)

4.01 Fractions (booklet p. 3)

WE DO examples (teacher)

- Column 1, E1: $\frac{4}{8} \rightarrow \frac{1}{2}$
- Column 1, E2: $\frac{6}{9} \rightarrow \frac{2}{3}$
- Column 1, E3: $2\frac{3}{4} \rightarrow \frac{11}{4}$
- Column 2, E1: $\frac{2}{3} = \frac{?}{9} \rightarrow 3 \times 3 = 9; 2 \times 3 = 6$
- Column 2, E2: $\frac{1}{2} = \frac{?}{16} \rightarrow 2 \times 8 = 16; 1 \times 8 = 8$
- Column 3, E1: $\frac{12}{20} \rightarrow$ HCF of 12 and 20 is 4; $12 \div 4 = 3, 20 \div 4 = 5; = \frac{3}{5}$
- Column 3, E2: $\frac{36}{42} \rightarrow$ HCF of 36 and 42 is 6; $36 \div 6 = 6, 42 \div 6 = 7; = \frac{6}{7}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------------|---------------------|--------------------|
| 1. $\frac{7}{10}$ | 15. $\frac{9}{2}$ | 32. 88 |
| 2. $\frac{1}{2}$ | 16. $\frac{13}{3}$ | 33. 5 |
| 3. $\frac{9}{10}$ | 17. $\frac{28}{5}$ | 34. 10 |
| 4. $\frac{2}{3}$ | 18. $\frac{13}{6}$ | 35. 6 |
| 5. $\frac{1}{2}$ | 19. $\frac{49}{10}$ | 36. 42 |
| 6. $\frac{2}{5}$ | 20. $\frac{5}{2}$ | 37. 9 |
| 7. $\frac{4}{5}$ | 21. $\frac{10}{3}$ | 38. 45 |
| 8. $\frac{3}{4}$ | 22. $\frac{8}{3}$ | 39. 3 |
| 9. $\frac{11}{12}$ | 23. $\frac{27}{4}$ | 40. 24 |
| 10. $\frac{1}{6}$ | 24. $\frac{16}{5}$ | 41. 4 |
| 11. $\frac{1}{3}$ | 25. $\frac{13}{4}$ | 42. 4 |
| 12. $\frac{7}{8}$ | 26. 6 | 43. 14 |
| 13. $\frac{22}{5}$ | 27. 24 | 44. 4 |
| 14. $\frac{17}{3}$ | 28. 6 | 45. 6 |
| | 29. 25 | 46. $\frac{8}{9}$ |
| | 30. 9 | 47. $\frac{1}{8}$ |
| | 31. 14 | 48. $\frac{6}{11}$ |
| | | 49. $\frac{4}{9}$ |
| | | 50. $\frac{7}{13}$ |
| | | 51. $\frac{3}{10}$ |

4.02 Ordering fractions (booklet p. 4)

WE DO examples (teacher)

- Column 1, E1: $\frac{5}{12}$ or $\frac{3}{12} \rightarrow \frac{5}{12}$
- Column 1, E2: $\frac{2}{10}$ or $\frac{9}{10} \rightarrow \frac{9}{10}$
- Column 2, E1: $\frac{1}{3}$ or $\frac{7}{8} \rightarrow \frac{8}{24}$ and $\frac{21}{24}$; larger: $\frac{7}{8}$
- Column 2, E2: $\frac{1}{10}$ or $\frac{7}{8} \rightarrow \frac{4}{40}$ and $\frac{35}{40}$; larger: $\frac{7}{8}$
- Column 3, E1: $\frac{1}{12}, \frac{1}{4}, \frac{3}{4} \rightarrow \frac{1}{12}, \frac{3}{12}, \frac{9}{12}; \frac{1}{12}, \frac{1}{4}, \frac{3}{4}$
- Column 3, E2: $\frac{2}{3}, \frac{1}{6}, \frac{5}{12} \rightarrow \frac{8}{12}, \frac{2}{12}, \frac{5}{12}; \frac{1}{6}, \frac{5}{12}, \frac{2}{3}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------------------|-------------------|--------------------|
| 1. $\frac{5}{8}$ | 5. $\frac{8}{9}$ | 9. $\frac{7}{10}$ |
| 2. $\frac{3}{4}$ | 6. $\frac{8}{9}$ | 10. $\frac{5}{8}$ |
| 3. $\frac{3}{4}$ | 7. $\frac{3}{5}$ | 11. $\frac{7}{11}$ |
| 4. $\frac{7}{8}$ | 8. $\frac{7}{10}$ | 12. $\frac{9}{12}$ |

- | | | |
|--------------------|--------------------|---|
| 13. $\frac{4}{9}$ | 27. $\frac{7}{9}$ | 41. $\frac{1}{3}$ |
| 14. $\frac{2}{3}$ | 28. $\frac{7}{12}$ | 42. $\frac{1}{2}$ |
| 15. $\frac{8}{9}$ | 29. $\frac{4}{7}$ | 43. $\frac{9}{10}$ |
| 16. $\frac{2}{3}$ | 30. $\frac{7}{10}$ | 44. $\frac{3}{4}$ |
| 17. $\frac{2}{7}$ | 31. $\frac{7}{8}$ | 45. $\frac{1}{2}$ |
| 18. $\frac{4}{5}$ | 32. $\frac{1}{2}$ | 46. $\frac{1}{3}, \frac{1}{2}, \frac{3}{4}$ |
| 19. $\frac{6}{10}$ | 33. $\frac{1}{2}$ | 47. $\frac{5}{12}, \frac{1}{2}, \frac{7}{12}$ |
| 20. $\frac{6}{7}$ | 34. $\frac{4}{5}$ | 48. $\frac{1}{6}, \frac{1}{2}, \frac{3}{4}$ |
| 21. $\frac{2}{5}$ | 35. $\frac{9}{10}$ | 49. $\frac{1}{4}, \frac{1}{2}, \frac{2}{3}$ |
| 22. $\frac{4}{8}$ | 36. $\frac{4}{5}$ | 50. $\frac{1}{4}, \frac{5}{6}, \frac{11}{12}$ |
| 23. $\frac{8}{12}$ | 37. $\frac{1}{4}$ | 51. $\frac{1}{6}, \frac{5}{12}, \frac{1}{2}$ |
| 24. $\frac{4}{5}$ | 38. $\frac{3}{4}$ | 52. $\frac{1}{3}, \frac{1}{2}, \frac{5}{6}$ |
| 25. $\frac{8}{12}$ | 39. $\frac{9}{10}$ | |
| 26. $\frac{7}{9}$ | 40. $\frac{3}{4}$ | |

4.03 Adding and subtracting fractions

(booklet p. 5)

WE DO examples (teacher)

- Column 1, E1: $\frac{4}{5} - \frac{1}{5} \rightarrow \frac{3}{5}$
- Column 1, E2: $\frac{2}{3} + \frac{2}{3} \rightarrow 1\frac{1}{3}$
- Column 2, E1: $\frac{1}{2} - \frac{1}{3} \rightarrow = \frac{3}{6} - \frac{2}{6}; = \frac{1}{6}$
- Column 2, E2: $\frac{2}{3} + \frac{1}{4} \rightarrow = \frac{8}{12} + \frac{3}{12}; = \frac{11}{12}$
- Column 3, E1: $\frac{1}{2} + \frac{1}{6} - \frac{1}{12} \rightarrow = \frac{6}{12} + \frac{2}{12} - \frac{1}{12}; = \frac{7}{12}; = \frac{7}{12}$
- Column 3, E2: $\frac{1}{2} + \frac{1}{3} - \frac{1}{6} \rightarrow = \frac{6}{12} + \frac{4}{12} - \frac{2}{12}; = \frac{8}{12}; = \frac{2}{3}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|---------------------|----------------------|----------------------|
| 1. 1 | 18. $\frac{8}{11}$ | 34. $\frac{13}{24}$ |
| 2. $1\frac{1}{5}$ | 19. $1\frac{1}{11}$ | 35. $1\frac{27}{40}$ |
| 3. $\frac{1}{5}$ | 20. 1 | 36. $\frac{3}{5}$ |
| 4. $\frac{3}{5}$ | 21. $\frac{1}{12}$ | 37. $1\frac{19}{30}$ |
| 5. 1 | 22. $\frac{3}{11}$ | 38. $\frac{7}{8}$ |
| 6. 1 | 23. $\frac{1}{11}$ | 39. $1\frac{11}{30}$ |
| 7. $\frac{6}{7}$ | 24. $\frac{2}{5}$ | 40. $\frac{1}{15}$ |
| 8. $1\frac{1}{3}$ | 25. $1\frac{3}{5}$ | 41. $1\frac{9}{40}$ |
| 9. $1\frac{2}{7}$ | 26. $\frac{9}{10}$ | 42. $\frac{17}{30}$ |
| 10. $1\frac{2}{7}$ | 27. $\frac{3}{4}$ | 43. $\frac{7}{24}$ |
| 11. $1\frac{4}{11}$ | 28. $\frac{1}{2}$ | 44. $1\frac{5}{12}$ |
| 12. $\frac{1}{4}$ | 29. $\frac{2}{3}$ | 45. $\frac{2}{5}$ |
| 13. $1\frac{3}{8}$ | 30. $1\frac{1}{2}$ | 46. $1\frac{3}{10}$ |
| 14. $\frac{2}{3}$ | 31. $\frac{3}{20}$ | 47. $\frac{5}{12}$ |
| 15. $\frac{1}{8}$ | 32. $1\frac{1}{15}$ | 48. $\frac{1}{2}$ |
| 16. $\frac{1}{3}$ | 33. $1\frac{13}{24}$ | 49. $\frac{1}{2}$ |
| 17. 1 | | |

50. $\frac{5}{12}$

51. $\frac{7}{12}$

4.04 Adding and subtracting mixed numerals (booklet p. 6)

WE DO examples (teacher)

- Column 1, E1: $1\frac{1}{4} + 5\frac{1}{4} \rightarrow 6\frac{1}{2}$
- Column 1, E2: $1\frac{2}{5} + 4\frac{2}{5} \rightarrow 5\frac{4}{5}$
- Column 2, E1: $5\frac{1}{3} - 3\frac{2}{3} \rightarrow = \frac{16}{3} - \frac{11}{3}; = \frac{5}{3} = 1\frac{2}{3}$
- Column 2, E2: $6\frac{1}{5} - 1\frac{4}{5} \rightarrow = \frac{31}{5} - \frac{9}{5}; = \frac{22}{5} = 4\frac{2}{5}$
- Column 3, E1: $4\frac{1}{4} + 2\frac{4}{5} \rightarrow = \frac{17}{4} + \frac{14}{5}; = \frac{85}{20} + \frac{56}{20}; = 7\frac{1}{20}$
- Column 3, E2: $2\frac{3}{4} + 1\frac{1}{3} \rightarrow = \frac{11}{4} + \frac{4}{3}; = \frac{33}{12} + \frac{16}{12}; = 4\frac{1}{12}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------------|---------------------|---------------------|
| 1. $6\frac{4}{7}$ | 18. $11\frac{6}{7}$ | 35. $1\frac{4}{5}$ |
| 2. $5\frac{1}{2}$ | 19. $11\frac{1}{2}$ | 36. $\frac{1}{3}$ |
| 3. $9\frac{2}{3}$ | 20. $9\frac{5}{9}$ | 37. $2\frac{3}{4}$ |
| 4. $7\frac{2}{7}$ | 21. $12\frac{2}{3}$ | 38. $\frac{4}{5}$ |
| 5. $5\frac{1}{2}$ | 22. $3\frac{7}{9}$ | 39. $2\frac{4}{5}$ |
| 6. $8\frac{4}{5}$ | 23. $5\frac{4}{7}$ | 40. $1\frac{1}{2}$ |
| 7. $11\frac{3}{5}$ | 24. $6\frac{8}{9}$ | 41. $1\frac{4}{5}$ |
| 8. $6\frac{5}{7}$ | 25. $10\frac{4}{5}$ | 42. $2\frac{3}{4}$ |
| 9. $5\frac{4}{5}$ | 26. $9\frac{4}{5}$ | 43. $1\frac{2}{3}$ |
| 10. $7\frac{4}{5}$ | 27. $7\frac{2}{3}$ | 44. $\frac{2}{3}$ |
| 11. $5\frac{1}{2}$ | 28. $6\frac{3}{5}$ | 45. $2\frac{1}{2}$ |
| 12. $5\frac{2}{3}$ | 29. $8\frac{3}{5}$ | 46. $5\frac{2}{3}$ |
| 13. $4\frac{2}{3}$ | 30. $2\frac{5}{7}$ | 47. $5\frac{1}{15}$ |
| 14. $3\frac{6}{7}$ | 31. $1\frac{2}{3}$ | 48. $6\frac{1}{2}$ |
| 15. $8\frac{3}{7}$ | 32. $2\frac{2}{3}$ | 49. $6\frac{4}{15}$ |
| 16. $2\frac{2}{5}$ | 33. $5\frac{4}{5}$ | 50. $7\frac{1}{6}$ |
| 17. $8\frac{5}{9}$ | 34. $4\frac{2}{3}$ | 51. $8\frac{4}{15}$ |

4.05 Fraction of a quantity (booklet p. 7)

WE DO examples (teacher)

- Column 1, E1: $\frac{1}{8}$ of 80 $\rightarrow 10$
- Column 1, E2: $\frac{1}{10}$ of 70 $\rightarrow 7$
- Column 2, E1: $\frac{3}{4}$ of 28 $\rightarrow 28 \div 4 = 7; 7 \times 3 = 21$
- Column 2, E2: $\frac{2}{3}$ of 21 $\rightarrow 21 \div 3 = 7; 7 \times 2 = 14$
- Column 3, E1: $\frac{1}{3}$ of a day, in hours $\rightarrow$ a day = 24 hours; $24 \div 3 = 8; 8 \times 1 = 8$ hours
- Column 3, E2: $\frac{3}{5}$ of an hour, in minutes $\rightarrow$ an hour = 60 minutes; $60 \div 5 = 12; 12 \times 3 = 36$ minutes

Questions (in order: Easy, Medium, Challenging)

- | | | |
|------|--------|--------|
| 1. 9 | 7. 3 | 13. 3 |
| 2. 8 | 8. 12 | 14. 12 |
| 3. 3 | 9. 2 | 15. 10 |
| 4. 4 | 10. 7 | 16. 6 |
| 5. 7 | 11. 2 | 17. 6 |
| 6. 2 | 12. 12 | 18. 4 |

19. 8

20. 6

21. 7

22. 11

23. 4

24. 8

25. 5

26. 8

27. 10

28. 12

29. 12

30. 80

31. 12

32. 21

33. 12

34. 6

35. 32

36. 55

37. 30

38. 9

39. 24

40. 12

41. 18

42. 16

43. 27

44. 24

45. 45

46. 12

47. 40

48. 10

49. 24

50. 5 minutes

51. 25 minutes

52. 24 minutes

53. 25 cents

54. 600 grams

55. 10 cents

4.06 Multiplying fractions (booklet p. 8)

WE DO examples (teacher)

- Column 1, E1: $\frac{8}{9} \times \frac{1}{2} \rightarrow \frac{4}{9}$
- Column 1, E2: $\frac{1}{2} \times \frac{4}{5} \rightarrow \frac{2}{5}$
- Column 2, E1: $\frac{2}{5} \times \frac{1}{14} \rightarrow = \frac{1}{5} \times \frac{1}{7}; = \frac{1}{35}$
- Column 2, E2: $\frac{4}{5} \times \frac{3}{14} \rightarrow = \frac{2}{5} \times \frac{3}{7}; = \frac{6}{35}$
- Column 3, E1: $1\frac{1}{3} \times 2\frac{3}{5} \rightarrow = \frac{4}{3} \times \frac{13}{5}; = \frac{52}{15}; = 3\frac{7}{15}$
- Column 3, E2: $1\frac{3}{4} \times 1\frac{2}{3} \rightarrow = \frac{7}{4} \times \frac{5}{3}; = \frac{35}{12}; = 2\frac{11}{12}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------------|--------------------|----------------------|
| 1. $\frac{1}{6}$ | 18. $\frac{1}{30}$ | 35. $\frac{4}{25}$ |
| 2. $\frac{1}{6}$ | 19. $\frac{3}{7}$ | 36. $\frac{2}{35}$ |
| 3. $\frac{4}{7}$ | 20. $\frac{4}{27}$ | 37. $\frac{1}{10}$ |
| 4. $\frac{1}{5}$ | 21. $\frac{1}{6}$ | 38. $\frac{1}{9}$ |
| 5. $\frac{1}{4}$ | 22. $\frac{1}{5}$ | 39. $\frac{9}{35}$ |
| 6. $\frac{3}{16}$ | 23. $\frac{1}{9}$ | 40. $\frac{2}{25}$ |
| 7. $\frac{1}{8}$ | 24. $\frac{1}{4}$ | 41. $\frac{3}{28}$ |
| 8. $\frac{8}{15}$ | 25. $\frac{1}{4}$ | 42. $\frac{2}{25}$ |
| 9. $\frac{1}{15}$ | 26. $\frac{1}{3}$ | 43. $\frac{1}{8}$ |
| 10. $\frac{1}{8}$ | 27. $\frac{3}{8}$ | 44. $\frac{3}{7}$ |
| 11. $\frac{2}{21}$ | 28. $\frac{3}{16}$ | 45. $\frac{1}{4}$ |
| 12. $\frac{2}{15}$ | 29. $\frac{2}{9}$ | 46. $\frac{2}{49}$ |
| 13. $\frac{1}{5}$ | 30. $\frac{7}{32}$ | 47. $4\frac{1}{5}$ |
| 14. $\frac{5}{18}$ | 31. $\frac{6}{49}$ | 48. $6\frac{13}{20}$ |
| 15. $\frac{1}{12}$ | 32. $\frac{1}{5}$ | 49. $1\frac{7}{9}$ |
| 16. $\frac{4}{25}$ | 33. $\frac{3}{35}$ | 50. $5\frac{7}{10}$ |
| 17. $\frac{1}{3}$ | 34. $\frac{6}{49}$ | 51. $7\frac{7}{10}$ |

4.07 Dividing fractions (booklet p. 9)

WE DO examples (teacher)

- Column 1, E1: $\frac{4}{8} \rightarrow \frac{8}{4}$
- Column 1, E2: $\frac{1}{7} \rightarrow 7$
- Column 2, E1: $\frac{5}{7} \div \frac{4}{5} \rightarrow = \frac{5}{7} \times \frac{5}{4}; = \frac{25}{28}$
- Column 2, E2: $\frac{4}{9} \div \frac{6}{7} \rightarrow = \frac{4}{9} \times \frac{7}{6}; = \frac{14}{27}$
- Column 3, E1: $2\frac{1}{5} \div 2\frac{1}{3} \rightarrow = \frac{11}{5} \div \frac{7}{3}; = \frac{11}{5} \times \frac{3}{7}; = \frac{33}{35}$

6. Column 3, E2: $3\frac{1}{4} \div 1\frac{3}{5} \rightarrow = \frac{13}{4} \div \frac{8}{5}; = \frac{13}{4} \times \frac{5}{8}; = 2\frac{1}{32}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|--------------------|--------------------|--------------------|
| 1. 2 | 18. $\frac{12}{7}$ | 34. $\frac{6}{7}$ |
| 2. $\frac{12}{11}$ | 19. $\frac{7}{5}$ | 35. $\frac{8}{9}$ |
| 3. $\frac{4}{9}$ | 20. $\frac{5}{9}$ | 36. $2\frac{1}{7}$ |
| 4. $\frac{2}{5}$ | 21. $\frac{11}{8}$ | 37. $\frac{5}{12}$ |
| 5. $\frac{9}{3}$ | 22. $\frac{12}{6}$ | 38. $\frac{5}{18}$ |
| 6. $\frac{10}{8}$ | 23. $\frac{7}{6}$ | 39. 2 |
| 7. $\frac{12}{5}$ | 24. $\frac{9}{6}$ | 40. $\frac{1}{2}$ |
| 8. $\frac{4}{11}$ | 25. $\frac{10}{2}$ | 41. $1\frac{3}{7}$ |
| 9. $\frac{11}{4}$ | 26. $\frac{4}{5}$ | 42. $\frac{2}{9}$ |
| 10. $\frac{8}{7}$ | 27. $\frac{8}{2}$ | 43. $1\frac{1}{3}$ |
| 11. $\frac{5}{11}$ | 28. $\frac{2}{6}$ | 44. $\frac{9}{10}$ |
| 12. 5 | 29. $\frac{4}{6}$ | 45. 2 |
| 13. $\frac{7}{3}$ | 30. $\frac{5}{10}$ | 46. 2 |
| 14. $\frac{10}{5}$ | 31. $\frac{2}{9}$ | 47. 1 |
| 15. $\frac{11}{6}$ | 32. $1\frac{7}{9}$ | 48. $2\frac{2}{3}$ |
| 16. $\frac{5}{4}$ | 33. $1\frac{1}{4}$ | 49. $3\frac{3}{4}$ |
| 17. $\frac{3}{10}$ | | 50. $\frac{9}{10}$ |
| | | 51. $3\frac{1}{2}$ |

4.08 Fraction problems (booklet p. 10)

WE DO examples (teacher)

- Column 1, E1: $\frac{2}{5}$ of a cake is sold $\rightarrow \frac{3}{5}$
- Column 1, E2: $\frac{2}{3}$ of a cake is sold $\rightarrow \frac{1}{3}$
- Column 2, E1: Sam had \$120 and spent $\frac{3}{10}$ of it. How much is left? $\rightarrow$ spent: $120 \div 10 \times 3 = \$36$; left: $120 - 36 = \$84$
- Column 2, E2: Sam had \$45 and spent $\frac{4}{5}$ of it. How much is left? $\rightarrow$ spent: $45 \div 5 \times 4 = \$36$; left: $45 - 36 = \$9$
- Column 3, E1: 12 students: $\frac{1}{3}$ walk to school and $\frac{1}{6}$ ride. The rest take the bus. How many take the bus? $\rightarrow$ walk: $12 \div 3 = 4$; ride: $12 \div 6 = 2$; bus: $12 - 4 - 2 = 6$
- Column 3, E2: 80 students: $\frac{1}{4}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus? $\rightarrow$ walk: $80 \div 4 = 20$; ride: $80 \div 5 = 16$; bus: $80 - 20 - 16 = 44$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------------|-------------------|-------------------|
| 1. $\frac{1}{3}$ | 10. $\frac{1}{6}$ | 19. $\frac{1}{2}$ |
| 2. $\frac{1}{12}$ | 11. $\frac{1}{8}$ | 20. $\frac{3}{5}$ |
| 3. $\frac{1}{12}$ | 12. $\frac{2}{3}$ | 21. $\frac{2}{5}$ |
| 4. $\frac{1}{4}$ | 13. $\frac{7}{8}$ | 22. \$49 |
| 5. $\frac{3}{5}$ | 14. $\frac{2}{3}$ | 23. \$16 |
| 6. $\frac{3}{4}$ | 15. $\frac{1}{2}$ | 24. \$12 |
| 7. $\frac{1}{2}$ | 16. $\frac{2}{3}$ | 25. \$33 |
| 8. $\frac{9}{10}$ | 17. $\frac{2}{5}$ | 26. \$84 |
| 9. $\frac{5}{6}$ | 18. $\frac{1}{2}$ | 27. \$18 |
| | | 28. \$27 |
| | | 29. 20 |
| | | 30. 3 |
| | | 31. 14 |
| | | 32. 15 |

4.09 Fractions on a calculator (booklet p. 11)

WE DO examples (teacher)

- Column 1, E1: $\frac{2}{10} + \frac{9}{19} \rightarrow \frac{64}{95}$
- Column 1, E2: $\frac{4}{14} + \frac{9}{12} \rightarrow 1\frac{1}{28}$
- Column 2, E1: $\frac{11}{14} \div \frac{3}{6} \rightarrow$ typed: $11/14 \div 3/6; = 1\frac{4}{7}$
- Column 2, E2: $\frac{10}{16} \div \frac{11}{13} \rightarrow$ typed: $10/16 \div 11/13; = \frac{65}{88}$
- Column 3, E1: $(\frac{1}{4} + \frac{3}{7}) \times \frac{6}{9} \rightarrow$ bracket: $\frac{19}{28}; \times \frac{6}{9} = \frac{19}{42}$
- Column 3, E2: $(\frac{2}{8} + \frac{1}{7}) \times \frac{2}{3} \rightarrow$ bracket: $\frac{11}{28}; \times \frac{2}{3} = \frac{11}{42}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|----------------------|----------------------|----------------------|
| 1. $\frac{20}{171}$ | 16. $\frac{3}{65}$ | 31. $3\frac{7}{38}$ |
| 2. $1\frac{3}{10}$ | 17. $1\frac{1}{18}$ | 32. $\frac{3}{4}$ |
| 3. $\frac{1}{8}$ | 18. $\frac{8}{15}$ | 33. $1\frac{1}{8}$ |
| 4. $\frac{3}{40}$ | 19. $\frac{65}{72}$ | 34. $1\frac{1}{2}$ |
| 5. $\frac{63}{152}$ | 20. $\frac{33}{272}$ | 35. $1\frac{1}{3}$ |
| 6. $\frac{13}{40}$ | 21. $\frac{23}{24}$ | 36. $\frac{9}{40}$ |
| 7. $1\frac{7}{95}$ | 22. $1\frac{35}{57}$ | 37. $2\frac{17}{20}$ |
| 8. $\frac{9}{10}$ | 23. $1\frac{32}{65}$ | 38. $\frac{11}{15}$ |
| 9. $\frac{161}{208}$ | 24. $\frac{21}{34}$ | 39. $\frac{49}{150}$ |
| 10. $\frac{2}{3}$ | 25. $\frac{41}{48}$ | 40. $1\frac{8}{27}$ |
| 11. $\frac{85}{112}$ | 26. $1\frac{17}{88}$ | 41. $\frac{25}{64}$ |
| 12. $1\frac{29}{70}$ | 27. $\frac{5}{18}$ | 42. $\frac{7}{54}$ |
| 13. $\frac{10}{33}$ | 28. $\frac{13}{15}$ | 43. $\frac{22}{27}$ |
| 14. $\frac{1}{4}$ | 29. $1\frac{4}{21}$ | 44. $\frac{37}{72}$ |
| 15. $1\frac{19}{91}$ | 30. $\frac{50}{153}$ | 45. $\frac{1}{8}$ |

4.10 Percentages, fractions and decimals

(booklet p. 12)

WE DO examples (teacher)

- Column 1, E1: 40% $\rightarrow \frac{2}{5}$
- Column 1, E2: 45% $\rightarrow \frac{9}{20}$
- Column 1, E3: 96% $\rightarrow 0.96$
- Column 2, E1: $\frac{3}{4} \rightarrow = \frac{75}{100}; = 75\%$
- Column 2, E2: $\frac{3}{10} \rightarrow = \frac{30}{100}; = 30\%$
- Column 3, E1: 0.6 $\rightarrow \times 100: 60\%; \frac{60}{100} = \frac{3}{5}$
- Column 3, E2: 0.75 $\rightarrow \times 100: 75\%; \frac{75}{100} = \frac{3}{4}$

Questions (in order: Easy, Medium, Challenging)

- | | | |
|-------------------|--------------------|----------|
| 1. $\frac{2}{25}$ | 8. $\frac{3}{4}$ | 17. 0.89 |
| 2. $\frac{7}{20}$ | 9. $\frac{3}{20}$ | 18. 0.66 |
| 3. $\frac{1}{10}$ | 10. $\frac{3}{10}$ | 19. 0.53 |
| 4. $\frac{1}{25}$ | 11. $\frac{7}{10}$ | 20. 1.46 |
| 5. $\frac{3}{5}$ | 12. $\frac{1}{5}$ | 21. 0.02 |
| 6. $\frac{4}{5}$ | 13. 0.38 | 22. 1.3 |
| 7. $\frac{1}{50}$ | 14. 0.16 | 23. 0.64 |
| | 15. 1.47 | 24. 0.84 |
| | 16. 0.9 | 25. 1.49 |
| | | 26. 90% |
| | | 27. 20% |
| | | 28. 25% |

29. 84%	38. 32%	46. 4%, $\frac{1}{25}$
30. 40%	39. 60%	47. 30%, $\frac{3}{10}$
31. 65%	40. 88%	48. 12%, $\frac{3}{25}$
32. 16%	41. 24%	49. 70%, $\frac{7}{10}$
33. 50%	42. 92%	50. 20%, $\frac{1}{5}$
34. 44%	43. 64%	51. 35%, $\frac{7}{20}$
35. 56%	44. 10%, $\frac{1}{10}$	
36. 8%	45. 25%, $\frac{1}{4}$	
37. 6%		

4.11 Percentage of a quantity (booklet p. 13)

WE DO examples (teacher)

- Column 1, E1: 50% of 142 → 71
- Column 1, E2: 25% of 32 → 8
- Column 2, E1: 15% of 160 → 10% = 16; 15% = 16 + 8 = 24
- Column 2, E2: 70% of 280 → 10% = 28; 70% = 28 × 7 = 196
- Column 3, E1: A \$200 jacket is 15% off. Find the sale price.
→ discount: 15% of 200 = \$30; sale price: 200 – 30 = \$170
- Column 3, E2: A \$40 jacket is 40% off. Find the sale price.
→ discount: 40% of 40 = \$16; sale price: 40 – 16 = \$24

Questions (in order: Easy, Medium, Challenging)

1. 9	18. 57	35. 162
2. 5	19. 73	36. 144
3. 85	20. 22	37. 30
4. 16	21. 22	38. 228
5. 8	22. 81	39. 18
6. 3	23. 15	40. 96
7. 25	24. 9	41. 90
8. 66	25. 3	42. 20
9. 37	26. 10	43. 96
10. 52	27. 11	44. 232
11. 8	28. 53	45. 200
12. 16	29. 49	46. 180
13. 29	30. 67	47. 88
14. 31	31. 15	48. 60
15. 4	32. 182	49. 156
16. 72	33. 204	50. 28
17. 2	34. 288	51. 16

52. 24	55. \$63	58. \$178.5
53. \$30	56. \$96	59. \$144
54. \$156	57. \$98	60. \$130

4.12 Expressing as a fraction or percentage (booklet p. 14)

WE DO examples (teacher)

- Column 1, E1: 2 out of 5 → 40%
- Column 1, E2: 3 out of 5 → 60%
- Column 2, E1: 8 out of 50 → $\frac{8}{50} = \frac{16}{100}$; = 16%
- Column 2, E2: 14 out of 20 → $\frac{14}{20} = \frac{70}{100}$; = 70%
- Column 3, E1: 250 g out of 2 kg → 2 kg = 2000 g; $\frac{250}{2000} \times 100\%$; = 12.5%
- Column 3, E2: 75c out of \$3 → \$3 = 300c; $\frac{75}{300} \times 100\%$; = 25%

Questions (in order: Easy, Medium, Challenging)

1. 44%	20. 45%	39. 65%
2. 20%	21. 5%	40. 6%
3. 90%	22. 18%	41. 8%
4. 40%	23. 30%	42. 30%
5. 32%	24. 50%	43. 20%
6. 25%	25. 80%	44. 95%
7. 70%	26. 60%	45. 10%
8. 10%	27. 70%	46. 96%
9. 10%	28. 55%	47. 24%
10. 50%	29. 95%	48. 15%
11. 20%	30. 8%	49. 85%
12. 75%	31. 25%	50. 40%
13. 52%	32. 75%	51. 12.5%
14. 85%	33. 56%	52. 25%
15. 50%	34. 20%	53. 25%
16. 76%	35. 50%	54. 37.5%
17. 26%	36. 50%	55. 20%
18. 36%	37. 56%	56. 12.5%
19. 10%	38. 60%	